Anatomy MCQ - Questions and Answers

CMU 1st Prof MBBS, Orbit Question Bank 18th ed. (2021-2025). In MFT questions each option is marked **T** (true) or **F** (false); correct-true options are in **green**. In SBA questions the best answer is in green with a tick mark (>>).

Answers follow standard textbooks (Snell, Gray's, Moore, BD Chaurasia, Langman, Junqueira, Ross); only the neuropore timing was checked online. Red italic notes mark where the Orbit key differs from textbooks or the question is ambiguous - confirm with your own book or teacher. Options were typed from a photographed scan, so a few spellings may differ from the original.

May 2025 - Paper I (Orbit p.76)

1. Fibrous joints
- a) Coronal suture **[T]**
- b) Temporomandibular joint [F]
- c) Inferior tibiofibular joint **[T]**
- d) Intervertebral joint [F]
- e) Manubriosternal joint [F]

2. Collagen fibers abundant in
- a) Bone **[T]**
- b) Lymph node [F]
- c) Lung [F]
- d) Tendon **[T]**
- e) Spleen [F]

3. Spermatogenesis
- a) Produce single haploid sperm [F]
- b) Starts before birth [F]
- c) Starts from puberty **[T]**
- d) Occurs in seminiferous tubule **[T]**
- e) Is called reduction division [F]

Note: Option e: reduction division (meiosis) is one step inside it, not the name of the whole process.

4. Middle ear communicates with
- a) Oral cavity [F]
- b) Nasal cavity [F]
- c) Inner ear **[T]**
- d) Nasopharynx **[T]**
- e) Mastoid antrum **[T]**

5. Extrapyramidal tracts
- a) Corticospinal [F]
- b) Reticulospinal **[T]**
- c) Rubrospinal **[T]**
- d) Spinotectal [F]
- e) Vestibulospinal **[T]**

6. Limbic system includes
- a) Hypothalamus **[T]**
- b) Hippocampus **[T]**
- c) Amygdala **[T]**
- d) Mammillary body **[T]**
- e) Claustrum [F]

7. Uniaxial joint
- a) Pivot **[T]**
- b) Ball and socket [F]
- c) Plane [F]
- d) Hinge **[T]**
- e) Ellipsoid [F]

8. MCA supplies
- a) Broca's area **[T]**
- b) Primary motor area **[T]**
- c) Occipital lobe [F]
- d) Basal ganglia **[T]**
- e) Internal capsule **[T]**

Note: Central (lenticulostriate) branches supply basal ganglia and much of the internal capsule. Orbit's key differs; textbooks vary slightly.

9. Central lymphoid organ
- a) Bone marrow **[T]**
- b) Lymph node [F]
- c) Palatine tonsil [F]
- d) Spleen [F]
- e) Thymus **[T]**

10. Pterion: meeting point of
- a) Frontal **[T]**
- b) Parietal **[T]**
- c) Temporal **[T]**
- d) Maxilla [F]
- e) Zygomatic [F]

Note: Also includes greater wing of sphenoid.

11. Muscles of facial expression
- a) Temporalis
- **>> b) Buccinator**
- c) Masseter
- d) Sternohyoid
- e) Omohyoid

12. Parasympathetic nucleus
- a) Vestibular nucleus
- b) Cochlear nucleus
- c) Red nucleus
- **>> d) Edinger-Westphal nucleus**
- e) Spinal nucleus

Note: Orbit marks b (cochlear nucleus) - wrong.

13. Endocrine gland
- a) Has duct
- **>> b) Has rich blood supply**
- c) Has rich nerve supply
- d) Secretes mucus fluid
- e) Secretes serous fluid

14. Retina
- a) Innermost layer contains pigment epithelium
- b) Bipolar cell form the optic nerve
- **>> c) Blind spot has no photoreceptors**
- d) Can regenerate
- e) Is avascular structure

Note: As printed, a is false (pigment epithelium is the OUTERMOST layer). Orbit marks a, so the question probably meant 'outermost'.

15. Cerebellum
- a) Parts of brain stem
- b) Controls opposite side of the body
- c) Directly controls the motor activity
- d) Arranged outer white matter and inner grey matter
- **>> e) Connected with the brain stem**

16. Mast cells
- **>> a) Participate in allergic reaction**
- b) Participate in phagocytosis
- c) Produce antibody
- d) Provide nutrition to epithelial tissue
- e) Synthesize ground substance

17. Skeletal muscle fiber
- a) Branching with intercalated discs
- b) Involuntary control with no striations
- **>> c) Multinucleated fibers with striations**
- d) Single central nucleus with no striations
- e) Spindle shaped fibers with central nuclei

18. Mature neurons are
- **>> a) Permanent cell**
- b) Undergo cell division
- c) Develops from all three germ layers
- d) Contain centriole
- e) Stable cell

19. Nerve passing through cavernous sinus
- **>> a) Abducent**
- b) Optic
- c) Facial
- d) Vestibulocochlear
- e) Mandibular

20. Basal nuclei
a) Medial geniculate body
b) Pulvinar
c) Red nucleus
d) Amygdaloid nucleus
>> e) Caudate nucleus

May 2025 - Paper II (Orbit p.77)

1. Female urinary bladder arteries
a) Superior vesical artery **[T]**
b) Inferior vesical artery [F]
c) Uterine artery **[T]**
d) Vaginal artery **[T]**
e) Ovarian artery [F]
Note: In females the vaginal artery replaces the inferior vesical artery.

2. Modifications of GI mucosa
a) Rugae **[T]**
b) Pyloric sphincter [F]
c) Microvilli **[T]**
d) Villi **[T]**
e) Plicae circulares **[T]**

3. Lateral branches of abdominal aorta supply
a) Gonads **[T]**
b) Kidneys **[T]**
c) Liver [F]
d) Large intestine [F]
e) The diaphragm **[T]**

4. Contents of adductor canal
a) Femoral artery **[T]**
b) Femoral vein **[T]**
c) Femoral nerve [F]
d) Saphenous nerve **[T]**
e) Great saphenous vein [F]
Note: Femoral artery, vein, saphenous nerve (and nerve to vastus medialis).

5. Dorsiflexion of foot
a) Tibialis anterior **[T]**
b) Extensor digitorum longus **[T]**
c) Flexor hallucis longus [F]
d) Peroneus tertius **[T]**
e) Peroneus brevis [F]
Note: Orbit marks EDL as F and flexor hallucis longus as T - wrong. FHL is a plantar flexor.

6. Posterior wall of axilla formed by
a) Serratus anterior [F]
b) Teres major **[T]**
c) Pectoralis major [F]
d) Subscapularis **[T]**
e) Latissimus dorsi **[T]**
Note: Orbit marks subscapularis as F - wrong. Posterior wall: subscapularis, teres major, latissimus dorsi.

7. Vena caval opening of diaphragm
a) Tendinous in nature **[T]**
b) Muscular in nature [F]
c) Closes during contraction of diaphragm [F]
d) Lies at 8th thoracic vertebra **[T]**
e) Transmits superior vena cava [F]
Note: Orbit marks c T - wrong; the opening is widened (not closed) in inspiration. It lies at T8.

8. Intercostal nerves
a) 11 in number [F]
b) 11 in pairs **[T]**
c) Supplies only thoracic wall [F]
d) Located at upper part of corresponding intercostal space **[T]**
e) Atypical intercostal nerves are more in number than typical ones **[T]**
Note: 11 pairs (so 'in number' option a is F).

9. Conducting system of heart
a) SA node **[T]**
b) AV node **[T]**
c) Papillary muscle [F]
d) AV bundle **[T]**
e) Musculi pectinati [F]

10. Muscle supplied by axillary nerve
a) Deltoid **[T]**
b) Teres major [F]
c) Teres minor **[T]**
d) Coracobrachialis [F]
e) Brachialis [F]
Note: Orbit marks teres major and coracobrachialis T - wrong (lower subscapular and musculocutaneous nerves).

11. Deep inguinal ring lies in layer
a) External oblique aponeurosis
b) Internal oblique aponeurosis
c) Transversus abdominis aponeurosis
>> d) Fascia transversalis
e) Fascia iliaca

12. Ligament preventing backward falling of trunk
>> a) Ilio femoral ligament
b) Ischio femoral ligament
c) Inguinal ligament
d) Pubofemoral ligament
e) Transverso acetabular ligament
Note: Orbit marks ischiofemoral. Standard texts (BD Chaurasia): iliofemoral (Y ligament) prevents the trunk falling backward.

13. Muscle connecting vertebral column with lower limb
a) Biceps femoris
b) Iliacus
c) Rectus femoris
>> d) Psoas major
e) Quadratus lumborum

14. Urethral fold fails to fuse
a) Epispadias
b) Ectopic vesicae
>> c) Hypospadias
d) Rectovesical fistula
e) Phimosis

15. Organ completely covered by peritoneum
a) Kidney
b) Pancreas
c) Ascending colon
>> d) Spleen
e) Rectum
Note: Except at the hilum.

16. Artery in superior mediastinum
a) Ascending aorta
b) Descending thoracic aorta
c) Internal thoracic artery
>> d) Left subclavian artery
e) Pulmonary trunk

17. Phrenic nerve supplies
>> a) Diaphragm
b) Heart
c) Intercostal muscles
d) Pulmonary pleura
e) Visceral pericardium
Note: Visceral pleura and visceral pericardium are not supplied by it.

18. Vertebrosternal ribs
>> a) 3, 4, 5
b) 6, 7, 8
c) 7, 8, 9
d) 9, 10, 11
e) 10, 11, 12
Note: Vertebrosternal ribs are 1-7; only option a lies fully within 1-7.

19. Submucous coat absent in
>> a) Gall bladder
b) Oesophagus
c) Duodenum
d) Vermiform appendix
e) Ileum

20. Multipennate muscle
a) Biceps brachii
b) Brachialis
c) Flexor digitorum profundus
d) Pectoralis major
>> e) Subscapularis

November 2025 - Paper I (Orbit p.78)

1. Pterion: meeting point of
a) Frontal **[T]**
b) Temporal **[T]**
c) Maxilla [F]
d) Zygomatic [F]
e) Parietal **[T]**
Note: Orbit marks zygomatic T and frontal F - wrong. Pterion = frontal, parietal, temporal, greater wing of sphenoid.

2. Simple columnar epithelium present in
a) Alveoli [F]
b) Duodenum **[T]**
c) Stomach **[T]**
d) Esophagus [F]
e) Tongue [F]

3. Extrapyramidal tracts
a) Corticospinal [F]
b) Reticulospinal **[T]**
c) Rubrospinal **[T]**
d) Spinotectal [F]
e) Vestibulospinal **[T]**

4. Refractive media of eyeball
a) Sclera [F]
b) Cornea **[T]**
c) Retina [F]
d) Iris [F]
e) Choroid [F]
Note: Cornea (aqueous, lens, vitreous are the others).

5. Mitosis
a) Occurs in somatic cell **[T]**
b) Occurs in mature germ cell [F]
c) Produces two cells of same chromosome number **[T]**
d) Is called reduction division [F]
e) Responsible for genetic variability [F]

6. Derivatives of mesoderm
a) Epidermis [F]
b) Cardiac muscle **[T]**
c) Lung [F]
d) Gall bladder [F]
e) Connective tissue **[T]**

7. Content of carotid sheath
a) Common carotid artery **[T]**
b) External carotid artery [F]
c) External jugular vein [F]
d) Internal jugular vein **[T]**
e) Vagus nerve **[T]**
Note: Common/internal carotid artery, IJV, vagus. External carotid is outside the sheath. Orbit's key differs.

8. Purely motor cranial nerve
a) Facial [F]
b) Glossopharyngeal [F]
c) Hypoglossal **[T]**
d) Trochlear **[T]**
e) Vestibulocochlear [F]

9. Central lymphoid organ
a) Bone marrow **[T]**
b) Lymph node [F]
c) Palatine tonsil [F]
d) Spleen [F]
e) Thymus **[T]**

10. Deep cerebellar nuclei
a) Caudate nucleus [F]
b) Dentate nucleus **[T]**
c) Globose **[T]**
d) Posteromedial nucleus [F]
e) Ventromedial nucleus [F]

11. Hard palate formed by
a) Ethmoid
b) Frontal
>> c) Maxilla
d) Zygomatic
e) Lacrimal

12. Muscle that opens laryngeal inlet
a) Aryepiglottic
b) Cricothyroid
c) Oblique arytenoid
d) Thyroarytenoid
>> e) Thyroepiglottic

13. Parasympathetic nucleus
a) Nucleus ambiguus
b) Vestibular
c) Mesencephalic
d) Red
>> e) Spinal
Note: Nucleus ambiguus is branchial motor; mesencephalic nucleus is sensory.

14. Effect of sympathetic system
a) Bronchoconstriction
b) Cardio inhibition
c) Coronary vasoconstriction
>> d) Decrease peristalsis
e) Decrease sweating

15. Reticular fiber found in
a) Bone
b) Tendon
>> c) Lymphoid organ
d) Cartilage
e) Muscle

16. Epithelial tissue
>> a) Is avascular
b) Develops from ectoderm only
c) Intercellular substance is maximum
d) Has nerve supply
e) Consists of cell, fiber and ground substance

17. Principal muscle of mastication
a) Buccinator
b) Digastric
c) Geniohyoid
>> d) Masseter
e) Mylohyoid

18. Cell in lacuna of Haversian system
a) Chondroblast
b) Osteoblast
c) Osteoclast
d) Osteoprogenitor
>> e) Osteocyte

19. Nerve through cavernous sinus
>> a) Abducent
b) Oculomotor
c) Optic
d) Ophthalmic
e) Trochlear
Note: Abducent runs freely through the sinus lumen; oculomotor, trochlear, ophthalmic and maxillary lie in its lateral wall.

20. Layer of scalp = 'safety valve' haematoma
a) Aponeurosis
b) Connective tissue
>> c) Loose areolar tissue
d) Pericranium
e) Skin

November 2025 - Paper II (Orbit p.79)

1. Branches of axillary artery
a) Superior thoracic artery **[T]**
b) Profunda brachii artery [F]
c) Subscapular artery **[T]**
d) Anterior circumflex humeral artery **[T]**
e) Ulnar artery [F]

2. Parietal pleura
a) Lines the thoracic wall and mediastinum **[T]**
b) Blood supply same as lung [F]
c) Develops from somatopleuric mesoderm **[T]**
d) Innervated by autonomic nerves [F]
e) Sensitive to pain **[T]**

3. Abduction of shoulder
a) Latissimus dorsi [F]
b) Pectoralis major [F]
c) Deltoid **[T]**
d) Supraspinatus **[T]**
e) Serratus anterior **[T]**
Note: Serratus anterior (with trapezius) rotates the scapula for abduction beyond 90 degrees. Orbit marks e F - debated.

4. Tributaries of coronary sinus
a) Great cardiac vein **[T]**
b) Anterior cardiac vein [F]
c) Venae cordis minimae [F]
d) Middle cardiac vein **[T]**
e) Right marginal vein [F]

5. Retroperitoneal organs
a) Stomach [F]
b) Ileum [F]
c) Pancreas **[T]**
d) Kidneys **[T]**
e) Ureters **[T]**

6. Mediastinal surface of right lung related to
a) Right atrium **[T]**
b) Left ventricle [F]
c) Arch of aorta [F]
d) Arch of azygos vein **[T]**
e) Right phrenic nerve [F]
Note: Right phrenic nerve lies on the mediastinal pleura but is not a groove/impression. Orbit also F.

7. Content of spermatic cord
a) Ductus deferens **[T]**
b) Inferior epigastric artery [F]
c) Testicular arteries **[T]**
d) Genital branch of genitofemoral nerve **[T]**
e) Femoral nerve [F]

8. Supplied by superior gluteal nerve
a) Gluteus medius **[T]**
b) Gluteus maximus [F]
c) Gluteus minimus **[T]**
d) Tensor fascia lata **[T]**
e) Gemellus superior [F]

9. Submucosal glands present in
a) Oesophagus **[T]**
b) Duodenum **[T]**
c) Stomach [F]
d) Ileum [F]
e) Appendix [F]
Note: Oesophageal glands proper lie in the submucosa (Junqueira). Orbit marks a F.

10. Derivatives of hindgut
a) Sigmoid colon **[T]**
b) Stomach [F]
c) Jejunum [F]
d) Appendix [F]
e) Rectum **[T]**

11. Winging of scapula caused by paralysis of
a) Subclavius muscle
b) Supraspinatus
c) Infraspinatus
d) Teres minor
>> e) Serratus anterior

12. Remnant of umbilical vein
a) Falciform ligament
b) Coronary ligament
>> c) Ligamentum teres hepatis
d) Ligamentum venosum
e) Triangular ligament

13. Branch arising from roots of brachial plexus
a) Median nerve
b) Radial nerve
c) Ulnar nerve
>> d) Long thoracic nerve
e) Axillary nerve

14. Diameter increased by pump-handle movement
a) Oblique diameter
>> b) Anteroposterior diameter
c) Transverse diameter
d) Vertical diameter
e) All diameters

15. Strongest ligament of hip joint
a) Ischiofemoral ligament
b) Capsular ligament
>> c) Iliofemoral ligament
d) Pubofemoral ligament
e) Ligamentum teres femoris

16. Conducting system supplied by left coronary artery
a) AV node
>> b) SA node
c) Purkinje fibers
d) Left branch of AV bundle
e) Right branch of AV bundle
Note: Weak question: SA node is from RCA in about 60%, left coronary in about 40%. b is the best option here.

17. Viscera related to right kidney
a) Pancreas
b) Spleen
c) Stomach
>> d) Liver
e) 3rd part of duodenum

18. Muscle forming inguinal ligament
a) Iliacus muscle
>> b) External oblique muscle
c) Transversus abdominis muscle
d) Internal oblique muscle
e) Rectus abdominis

19. Primary support of uterus
>> a) Pelvic diaphragm
b) Broad ligament
c) Uterovesical fold of peritoneum
d) Rectovaginal fold of peritoneum
e) Rectum

20. Keystone of medial longitudinal arch
a) Calcaneus
b) Navicular
c) Medial cuneiform
>> d) Talus
e) 1st metatarsal

May 2024 - Paper I (Orbit p.80)

1. Carotid sheath contains
a) Internal jugular vein **[T]**
b) External jugular vein [F]
c) Common carotid artery **[T]**
d) External carotid artery [F]
e) Internal carotid artery **[T]**

2. Reticular fiber found in
a) Spleen **[T]**
b) Lymph node **[T]**
c) Tendon [F]
d) Ligament [F]
e) Lung [F]

3. Skeletal muscle fiber
a) Are spindle in shape [F]
b) Has multiple nuclei **[T]**
c) Innervated by autonomic nerve [F]
d) Voluntary in function **[T]**
e) Develops from paraxial mesoderm **[T]**
Note: Orbit marks e F; limb and trunk skeletal muscle arises from somites (paraxial mesoderm).

4. Structural abnormalities of chromosome
a) Turner syndrome [F]
b) Deletion **[T]**
c) Ring chromosome **[T]**
d) Translocation **[T]**
e) Trisomy [F]
Note: Ring chromosome is structural; Orbit marks it F. Turner and trisomy are numerical.

5. Fibrous joints
a) Coronal suture **[T]**
b) Temporomandibular joint [F]
c) Inferior tibiofibular joint **[T]**
d) Intervertebral joint [F]
e) Manubriosternal joint [F]

6. Commissural fibers
a) Internal capsule [F]
b) Corpus callosum **[T]**
c) Superior longitudinal [F]
d) Optic radiation [F]
e) Anterior commissure **[T]**

7. Nasopharynx communicates with
a) Nasal cavity **[T]**
b) Middle ear cavity **[T]**
c) Oropharynx **[T]**
d) Oral cavity [F]
e) Larynx [F]
Note: Orbit marks b F, but it communicates with the middle ear via the auditory tube.

8. Extrapyramidal tracts
a) Corticospinal [F]
b) Reticulospinal **[T]**
c) Rubrospinal **[T]**
d) Spinotectal [F]
e) Vestibulospinal **[T]**

9. Endodermal derivatives
a) Thyroid gland **[T]**
b) Pancreas **[T]**
c) Parathyroid gland **[T]**
d) Sweat gland [F]
e) Adrenal gland [F]

10. Nasal septum formed by
a) Ethmoid bone **[T]**
b) Occipital bone [F]
c) Septal cartilage **[T]**
d) Temporal bone [F]
e) Vomer **[T]**

11. Avascular structure in eyeball
a) Retina
>> b) Lens
c) Sclera
d) Iris
e) Choroid

12. Artery wall thicker than vein due to
a) Absence of valve
b) Presence of internal and external elastic lamina
c) Presence of oxygenated blood
d) Presence of vasa vasorum
>> e) Well developed tunica media

13. Descending tract
a) Fasciculus gracilis
b) Spinotectal
c) Dorsal spinocerebellar
>> d) Tectospinal
e) Spinoolivary

14. Effect of sympathetic system
a) Bronchoconstriction
b) Cardio inhibition
c) Coronary vasoconstriction
>> d) Decrease peristalsis
e) Decrease sweating

15. Nissl granules present in
a) Ependymal cell
b) Microglia
c) Oligodendrocyte
d) Protoplasmic astrocyte
>> e) Pyramidal cell

16. Parasympathetic fibers present in
a) Optic nerve
>> b) Oculomotor nerve
c) Trigeminal nerve
d) Accessory nerve
e) Hypoglossal nerve

17. Hard palate formed by
a) Ethmoid
b) Frontal
>> c) Maxilla
d) Zygomatic
e) Lacrimal

18. Cell in lacuna of Haversian system
a) Chondroblast
b) Osteoblast
c) Osteoclast
d) Osteoprogenitor
>> e) Osteocyte

19. Multipolar neuron
a) Olfactory cell
>> b) Purkinje cell
c) Dorsal root ganglia
d) Trigeminal ganglia
e) Vestibular ganglion

20. Derivative of pharyngeal pouch
a) Thyroid gland
>> b) Thymus
c) Parotid gland
d) External auditory meatus
e) Ear ossicles

May 2024 - Paper II (Orbit p.81)

1. Structure crossed by root of mesentery
a) 2nd part of the duodenum [F]
b) Right psoas major **[T]**
c) Left ureter [F]
d) Abdominal aorta **[T]**
e) Left psoas major [F]
Note: Root crosses 3rd part of duodenum, aorta, IVC, right ureter, right psoas major. Orbit marks b F.

2. Veins ending into coronary sinus
a) Great cardiac vein **[T]**
b) Middle cardiac vein **[T]**
c) Anterior cardiac vein [F]
d) Small cardiac vein **[T]**
e) Venae cordis minimae [F]

3. Respiratory zone of bronchial tree
a) Conducts air from nose to alveoli [F]
b) Begins from respiratory bronchiole **[T]**
c) Has abundant goblet cells in its mucosa [F]
d) Supplied by branches of pulmonary artery **[T]**
e) Develops from ectoderm [F]
Note: Begins at respiratory bronchiole; alveolar capillaries come from the pulmonary artery. Weakly worded question.

4. Flexor of elbow joint
a) Brachialis **[T]**
b) Triceps brachii [F]
c) Coracobrachialis [F]
d) Biceps brachii **[T]**
e) Supraspinatus [F]

5. Nerve related to humerus
a) Median nerve [F]
b) Ulnar nerve **[T]**
c) Axillary nerve **[T]**
d) Radial nerve **[T]**
e) Musculocutaneous nerve [F]
Note: Ulnar nerve touches the back of the medial epicondyle. Orbit's key differs (marks e T).

6. Quadriceps femoris
a) Contain three muscles [F]
b) Powerful flexor of knee [F]
c) Is supplied by the tibial nerve [F]
d) Its tendon forms the ligamentum patellae **[T]**
e) Injury to femoral nerve causes its paralysis **[T]**

7. Histological features of duodenum
a) Serous cells [F]
b) Brunner's gland **[T]**
c) Simple squamous epithelium [F]
d) Chief cells [F]
e) Crypts of Lieberkuhn **[T]**

8. Contents of rectus sheath
a) Rectus abdominis **[T]**
b) Inferior epigastric artery **[T]**
c) Musculophrenic artery [F]
d) 6th intercostal nerve [F]
e) Subcostal nerve **[T]**

9. Ureter
a) 25 cm in length **[T]**
b) Has got four constrictions [F]
c) Develops from endoderm [F]
d) Lined by simple cuboidal epithelium [F]
e) Is retroperitoneal **[T]**
Note: Ureter develops from the ureteric bud (mesoderm), not endoderm; it has three constrictions; lined by transitional epithelium. Orbit marks c T.

10. Femoral artery
a) Passes through deep inguinal ring [F]
b) Content of femoral sheath **[T]**
c) Is a muscular artery **[T]**
d) Clinically not important [F]
e) Branch of internal iliac artery [F]

11. Groove behind hilum of right lung lodges
a) Descending thoracic aorta
b) Azygos vein
c) Superior pulmonary vein
>> d) Esophagus
e) Trachea

12. Most common position of vermiform appendix
a) Retro-ileal
>> b) Retro-caecal
c) Pelvic
d) Anterior to terminal ileum
e) Subcaecal

13. Most common site of clavicle fracture
a) Medial end
b) Lateral end
c) Midpoint
>> d) Junction of the medial two third and lateral one third
e) Junction of the lateral two third and medial one third

14. Transpyloric plane passes through
a) The spleen
b) The third part of the duodenum
c) The tail of the pancreas
>> d) The origin of the superior mesenteric artery
e) The neck of the gallbladder

15. Hepatic lobule
a) Hepatic lamina extends from periphery to center
b) Central vein has Kupffer cells in its wall
>> c) Portal triad lies at the corners of lobule
d) Sinusoids lie between two adjacent lobules
e) Is based on oxygenation of liver parenchyma

16. Lymphatics from nipple and areola drain mainly into
>> a) Pectoral group of axillary lymph node
b) Subscapular group of axillary lymph node
c) Central group of axillary lymph node
d) Apical group of axillary lymph node
e) Parasternal lymph group

17. Back of medial epicondyle related to
a) Radial nerve
b) Axillary nerve
>> c) Ulnar nerve
d) Median nerve
e) None of the above

18. Paralysis of which nerve causes foot drop
a) Femoral
b) Tibial
c) Superficial peroneal
>> d) Deep peroneal
e) Obturator
Note: Common peroneal nerve injury is the classic cause; of the options, deep peroneal.

19. Pleural reflection at mid-axillary line
a) 6th
b) 8th
c) 9th
>> d) 10th
e) 12th
Note: Pleura: 8th rib (midclavicular), 10th (midaxillary), 12th (paravertebral).

20. Anterior relation of sacroiliac joint
a) Femoral nerve
b) Sciatic nerve
>> c) Obturator nerve
d) Round ligament
e) Ureter

November 2024 - Paper I (Orbit p.82)

1. Paired dural venous sinuses
a) Cavernous **[T]**
b) Occipital [F]
c) Straight [F]
d) Superior sagittal [F]
e) Transverse **[T]**

2. Avascular structures
a) Bone [F]
b) Cornea **[T]**
c) Dermis [F]
d) Epidermis **[T]**
e) Hair **[T]**
Note: Orbit marks dermis T - wrong, the dermis is vascular.

3. Hyaline cartilage present in
a) Trachea **[T]**
b) Epiglottis [F]
c) Costal cartilage **[T]**
d) Menisci of knee [F]
e) External ear [F]

4. Peripheral lymphoid organ
a) Bone marrow [F]
b) Lymph node **[T]**
c) Palatine tonsil **[T]**
d) Thymus [F]
e) Spleen **[T]**
Note: The spleen is a peripheral lymphoid organ; Orbit marks it F.

5. Hard palate formed by
a) Maxilla **[T]**
b) Palatine bone **[T]**
c) Nasal bone [F]
d) Ethmoid bone [F]
e) Lacrimal bone [F]

6. Hind brain includes
a) Cerebellum **[T]**
b) Cerebrum [F]
c) Medulla oblongata **[T]**
d) Mid brain [F]
e) Pons **[T]**
Note: Pons belongs to the hindbrain; Orbit marks it F.

7. Superior orbital fissure transmits
a) Abducent nerve **[T]**
b) Maxillary nerve [F]
c) Middle meningeal artery [F]
d) Nasociliary nerve **[T]**
e) Oculomotor nerve **[T]**

8. Spinal cord
a) Contains multipolar neuron in anterior horn **[T]**
b) Contains preganglionic sympathetic neuron in thoracic region **[T]**
c) Develops from surface ectoderm [F]
d) Is covered by endosteal dura mater [F]
e) Is supplied by vertebral artery **[T]**

9. Meiosis cell division
a) Occurs in mature germ cell [F]
b) Occurs in immature germ cell **[T]**
c) Is called reduction division **[T]**
d) Produce two daughter cells [F]
e) Causes genetic variation **[T]**
Note: Meiosis occurs in primary spermatocytes/oocytes (immature germ cells); mature gametes do not divide. Orbit marks a T, b F - I disagree.

10. Deep cerebellar nucleus
a) Caudate nucleus [F]
b) Dentate nucleus **[T]**
c) Globose **[T]**
d) Posteromedial nucleus [F]
e) Ventromedial nucleus [F]

11. Safety muscle of tongue
a) Palatoglossus
>> b) Genioglossus
c) Hyoglossus
d) Styloglossus
e) Geniohyoid

12. Muscle that opens laryngeal inlet
a) Aryepiglottic
b) Cricothyroid
c) Oblique arytenoid
d) Thyroarytenoid
>> e) Thyroepiglottic

13. Parasympathetic nucleus
>> a) Dorsal nucleus
b) Vestibular nucleus
c) Mesencephalic nucleus
d) Red nucleus
e) Spinal nucleus

14. Effect of sympathetic system
a) Bronchoconstriction
b) Cardio inhibition
c) Coronary vasoconstriction
>> d) Decrease peristalsis
e) Decrease sweating

15. Bipolar neuron
a) Pyramidal cell
b) Purkinje cell
>> c) Olfactory cell
d) Anterior horn cell
e) Stellate cell

16. Basal nuclei
a) Medial geniculate body
b) Pulvinar
c) Red nucleus
>> d) Amygdaloid nucleus
e) Paraventricular nucleus

17. Principal muscle of mastication
a) Buccinator
b) Digastric
c) Geniohyoid
>> d) Masseter
e) Mylohyoid

18. Cell in lacuna of Haversian system
a) Chondroblast
b) Osteoblast
c) Osteoclast
d) Osteoprogenitor
>> e) Osteocyte

19. Primary cartilaginous joint
a) Pubic symphysis
b) Sacroiliac joint
c) Lumbosacral joint
>> d) First sternocostal joint
e) Temporomandibular

20. Characteristics of epithelium tissue
a) Has blood supply [F]
b) Has nerve supply **[T]**
c) Develops from all three germ layers **[T]**
d) Composed of fiber and matrix [F]
e) Intercellular substance is maximum [F]

November 2024 - Paper II (Orbit p.83)

1. Opening of right atrium
a) Superior vena cava **[T]**
b) Coronary sinus **[T]**
c) Anterior cardiac vein **[T]**
d) Pulmonary vein [F]
e) Ascending aorta [F]

2. Carpal tunnel contains
a) Flexor pollicis longus tendon **[T]**
b) Tendon of flexor digitorum superficialis **[T]**
c) Flexor carpi radialis [F]
d) Median nerve **[T]**
e) Ulnar artery [F]

3. Great saphenous vein
a) Connected with deep vein through perforator **[T]**
b) Site for venesection **[T]**
c) Begins at lateral side of foot [F]
d) Devoid of valves [F]
e) Passes behind the medial malleolus [F]
Note: It begins on the MEDIAL side of the foot, has valves, and passes in FRONT of the medial malleolus (Orbit marks e T).

4. Superior vena cava
a) Is located in posterior mediastinum [F]
b) Is a large vein **[T]**
c) Is formed by union of right and left subclavian vein [F]
d) Terminates in smooth part of right atrium **[T]**
e) Brings blood to heart from above the diaphragm **[T]**
Note: SVC is formed by the right and left BRACHIOCEPHALIC veins.

5. Alveoli
a) Lined by simple squamous epithelium **[T]**
b) Designed for air conduction [F]
c) Surrounded by pulmonary capillaries **[T]**
d) Develop from endoderm of foregut **[T]**
e) Non-collapsible [F]

6. Muscles that dorsiflex the foot
a) Tibialis anterior **[T]**
b) Flexor digitorum longus [F]
c) Extensor hallucis longus **[T]**
d) Peroneus tertius **[T]**
e) Tibialis posterior [F]

7. Intracapsular structure of knee
a) Cruciate ligament **[T]**
b) Menisci **[T]**
c) Popliteal tendon **[T]**
d) Patellar ligament [F]
e) Synovial membrane [F]
Note: Popliteus tendon is intracapsular but extrasynovial. Orbit's key varies between years for the synovial membrane.

8. Extra-hepatic biliary apparatus
a) Hepatic duct **[T]**
b) Bile canaliculi [F]
c) Cystic duct **[T]**
d) Bile duct **[T]**
e) Bile ductule [F]

9. Pancreas
a) Is an endocrine gland **[T]**
b) Lies in the epigastrium **[T]**
c) Is an intra-peritoneal structure [F]
d) Alpha cells secrete insulin [F]
e) Portal vein is formed behind its neck **[T]**
Note: Alpha cells secrete glucagon (beta cells secrete insulin). Pancreas is mostly retroperitoneal.

10. Anterior compartment of leg
a) Tibialis anterior **[T]**
b) Extensor hallucis longus **[T]**
c) Flexor digitorum longus [F]
d) Peroneus tertius **[T]**
e) Peroneus brevis [F]
Note: Flexor digitorum longus is in the posterior compartment; Orbit marks c T.

11. Strongest ligament of upper limb
a) Consoclavicular ligament (sic)
>> b) Coracoclavicular ligament
c) Suprascapular ligament
d) Glenohumeral ligament
e) Transverse humeral ligament

12. Strong supinator of forearm
a) Supinator
>> b) Biceps brachii
c) Brachioradialis
d) Brachialis
e) Coracobrachialis

13. Jejunum and ileum develop from
a) Foregut
>> b) Midgut
c) Hind gut
d) Cloaca
e) Pharyngeal gut

14. Parietal pleura
a) Supplied by phrenic nerve only
>> b) Pain sensitive
c) Not present over apex of lung
d) Derived from ectoderm
e) Lined by endothelium

15. NOT a content of femoral sheath
a) Femoral artery
b) Femoral vein
c) Femoral branch of genitofemoral nerve
>> d) Femoral nerve
e) Lymph node of Cloquet

16. Transpyloric plane passes through
a) T2-L1
>> b) L1-L2
c) T10
d) L5-S1
e) L2-L3
Note: The plane lies at L1 (some texts say lower L1/upper L2); L1-L2 is the closest option.

17. Referred pain of cholecystitis felt in
a) Tip of ninth costal cartilage
b) Around umbilicus
>> c) Tip of right shoulder
d) Right iliac fossa
e) Epigastrium

18. Least dilatable part of urethra
a) Pre-prostatic
b) Prostatic
>> c) Membranous
d) Spongy
e) Penile

19. Failure of testis to descend
a) Hypospadias
>> b) Cryptorchidism
c) Epispadias
d) Varicocele
e) Ectopic testis

20. Structure crossing floor of ovarian fossa
a) External iliac artery
b) Ovarian vessels
>> c) Obturator nerve
d) Pudendal nerve
e) Umbilical artery (printed 'Urine artery')

May 2023 - Paper I (old curriculum) (Orbit p.84)

1. Membranous organelles
a) Centrosome [F]
b) Chromosome [F]
c) Lysosome **[T]**
d) Mitochondria **[T]**
e) Ribosome [F]
Note: Ribosome is non-membranous; Orbit marks e T.

2. Mitotic cell division occurs in
a) Primary oocyte [F]
b) Primary spermatocyte [F]
c) Somatic cells **[T]**
d) Spermatozoa [F]
e) Zygote **[T]**

3. Simple squamous epithelium present in
a) Blood vessels **[T]**
b) Lung alveoli **[T]**
c) Pleura **[T]**
d) Tongue [F]
e) Urinary bladder [F]

4. Surface ectodermal derivatives
a) Bone [F]
b) Dermis [F]
c) Hair **[T]**
d) Sweat gland **[T]**
e) Thyroid gland [F]

5. Elastic cartilage present in
a) Auditory tube **[T]**
b) Epiglottis **[T]**
c) Intervertebral disc [F]
d) Thyroid cartilage [F]
e) Trachea [F]
Note: Auditory tube cartilage: textbooks vary; epiglottis is certain.

6. Secondary cartilaginous joint
a) Intervertebral joint **[T]**
b) Knee joint [F]
c) Manubriosternal joint **[T]**
d) Pubic symphysis **[T]**
e) Temporomandibular joint [F]

7. Avascular structures
a) Bone [F]
b) Cartilage **[T]**
c) Cornea **[T]**
d) Dermis [F]
e) Hair **[T]**

8. Surface modifications of cell
a) Cilia **[T]**
b) Flagella **[T]**
c) Microvilli **[T]**
d) Papillae [F]
e) Villi [F]
Note: Villi and papillae are tissue-level, not cell-level, modifications.

9. Peripheral lymphoid organ
a) Bone marrow [F]
b) Lymph node **[T]**
c) Palatine tonsil **[T]**
d) Spleen **[T]**
e) Thymus [F]

10. Connective tissue cell
a) Ependymal cell [F]
b) Fibroblast **[T]**
c) Mast cell **[T]**
d) Plasma cell **[T]**
e) Schwann cell [F]

11. Jugular foramen transmits
a) External jugular vein [F]
b) Glossopharyngeal nerve **[T]**
c) Internal jugular vein **[T]**
d) Mandibular nerve [F]
e) Vagus nerve **[T]**
Note: Also accessory nerve.

12. Muscles of mastication
a) Lateral pterygoid **[T]**
b) Masseter **[T]**
c) Stylohyoid [F]
d) Temporalis **[T]**
e) Tensor palati [F]

13. Motor cranial nerve
a) Abducens **[T]**
b) Glossopharyngeal [F]
c) Oculomotor **[T]**
d) Optic [F]
e) Trigeminal [F]

14. Derivatives of 2nd pharyngeal arch
a) Frontalis **[T]**
b) Mylohyoid muscle [F]
c) Orbicularis oris muscle **[T]**
d) Stylohyoid muscle **[T]**
e) Temporalis muscle [F]

15. Parotid gland
a) Contains facial nerve **[T]**
b) Is an exocrine gland **[T]**
c) Not covered by deep fascia [F]
d) Opens on the floor of oral cavity [F]
e) Secretes serous fluid **[T]**
Note: It is an exocrine gland and is enclosed by parotid fascia; the duct opens in the vestibule opposite the upper 2nd molar. Orbit marks b F, c T, d T.

16. Hind brain formed by
a) Cerebellum **[T]**
b) Hypothalamus [F]
c) Medulla oblongata **[T]**
d) Midbrain [F]
e) Thalamus [F]

17. Basal ganglia includes
a) Caudate nucleus **[T]**
b) Corpus callosum [F]
c) Globus pallidus **[T]**
d) Mesencephalic nucleus [F]
e) Putamen **[T]**

18. Deep cerebellar nuclei
a) Dentate nucleus **[T]**
b) Emboliform nucleus **[T]**
c) Fastigial nucleus **[T]**
d) Lentiform nucleus [F]
e) Red nucleus [F]

19. Occipital lobe of cerebrum
a) Contains auditory area [F]
b) Contains visual area **[T]**
c) Is a part of hind brain [F]
d) Situated below the tentorium cerebelli [F]
e) Supplied by posterior cerebral artery **[T]**
Note: It lies above (not below) the tentorium cerebelli.

20. Refractive media of eyeball
a) Cornea **[T]**
b) Iris [F]
c) Lens **[T]**
d) Sclera [F]
e) Vitreous humour **[T]**

November 2023 - Paper I (Orbit p.85)

1. Muscles of soft palate
a) Hyoglossus [F]
b) Palatoglossus **[T]**
c) Styloglossus [F]
d) Tensor tympani [F]
e) Tensor palati **[T]**
Note: Tensor tympani belongs to the middle ear. Orbit marks d T and e F.

2. Simple squamous epithelium present in
a) Alveoli **[T]**
b) Blood vessel **[T]**
c) Esophagus [F]
d) Peritoneum **[T]**
e) Tongue [F]
Note: Peritoneum = mesothelium (simple squamous); Orbit marks it F.

3. Superior orbital fissure transmits
a) Abducent nerve **[T]**
b) Oculomotor nerve **[T]**
c) Ophthalmic nerve **[T]**
d) Optic nerve [F]
e) Trochlear nerve **[T]**
Note: The optic nerve passes through the optic canal; Orbit marks d T.

4. Derivative of neural crest cell
a) Adrenal cortex [F]
b) Adrenal medulla **[T]**
c) Dorsal root ganglia **[T]**
d) Epidermis [F]
e) Schwann cell **[T]**

5. Miniature long bone
a) Clavicle [F]
b) Cuboid [F]
c) Metatarsal bone **[T]**
d) Phalanges **[T]**
e) Talus [F]

6. Derivatives of pharyngeal pouches
a) Internal ear [F]
b) Palatine tonsil **[T]**
c) Thymus **[T]**
d) Tongue [F]
e) Thyroid gland [F]

7. Part of basal ganglia
a) Caudate nucleus **[T]**
b) Dentate nucleus [F]
c) Globus pallidus **[T]**
d) Putamen **[T]**
e) Red nucleus [F]
Note: Dentate nucleus is cerebellar; Orbit marks it T.

8. Sensory cranial nerve
a) Facial [F]
b) Glossopharyngeal [F]
c) Hypoglossal [F]
d) Optic **[T]**
e) Vestibulocochlear **[T]**

9. Structures within parotid gland
a) Facial nerve **[T]**
b) Auriculotemporal nerve **[T]**
c) Glossopharyngeal nerve [F]
d) External carotid artery **[T]**
e) Internal carotid artery [F]
Note: Glossopharyngeal fibers reach the gland via lesser petrosal - otic ganglion - auriculotemporal nerve; the nerve itself is not within the gland.

10. Visual pathway formed by
a) Cornea [F]
b) Optic chiasma **[T]**
c) Optic tract **[T]**
d) Medial geniculate body [F]
e) Mamillary body [F]
Note: Medial geniculate body is auditory (lateral geniculate is visual). Cornea is not part of the pathway proper; Orbit marks it T.

11. Extra-ocular muscle supplied by abducent nerve
a) Inferior rectus
b) Inferior oblique
c) Medial rectus
>> d) Lateral rectus
e) Superior oblique

12. Nucleus of paleocerebellum
a) Dentate
b) Emboliform
c) Fastigial
d) Globose
>> e) Globose and emboliform
Note: Paleocerebellum (spinocerebellum) -> globose + emboliform; fastigial -> archicerebellum. Orbit marks c.

13. Laryngeal cartilage made of both hyaline and elastic
>> a) Arytenoid
b) Corniculate
c) Cricoid
d) Epiglottis
e) Thyroid
Note: Arytenoid: base hyaline, apex elastic. Epiglottis is elastic only.

14. Betz cell present in
a) External granular layer
b) External pyramidal layer
c) Internal granular layer
>> d) Internal pyramidal layer
e) Multiform layer

15. Terminal branch of internal carotid artery
a) Anterior choroidal artery
>> b) Anterior cerebral artery
c) Anterior communicating artery
d) Ophthalmic artery
e) Posterior communicating artery
Note: Middle cerebral is the other terminal branch.

16. Pseudounipolar neuron located at
a) Mesencephalic neuron of trigeminal nerve
b) Trigeminal ganglion
>> c) Dorsal root ganglion
d) Pyramidal cells of cerebrum
e) Retina
Note: Trigeminal ganglion cells are also pseudounipolar, so b is arguable; c is the textbook example.

17. Posterior neuropore closed at
a) 15th day
b) 20th day
c) 22nd day
d) 25th day
>> e) 28th day
Note: Langman: anterior neuropore ~day 25, posterior ~day 27-28. Orbit marks 25th day.

18. Safety muscle of larynx
a) Aryepiglottic
b) Lateral cricoarytenoid
c) Oblique arytenoid
>> d) Posterior cricoarytenoid
e) Thyroarytenoid

19. Unique feature of cervical vertebra
a) Bifid spine
>> b) Foramen transversarium
c) Triangular vertebral foramen
d) Long lamina
e) Small body

20. DNA replication occurs in
a) G0 phase
b) G1 phase
c) G2 phase
d) M phase
>> e) S phase

November 2023 - Paper II (Orbit p.86)

1. Contents of middle mediastinum
a) Pericardium **[T]**
b) Arch of aorta [F]
c) Phrenic nerve **[T]**
d) Pulmonary nerve (sic) [F]
e) Vagus nerve [F]
Note: Printed 'pulmonary nerve'; if 'pulmonary trunk' was meant, it is also T.

2. Carpal tunnel transmits
a) Median nerve **[T]**
b) Ulnar nerve [F]
c) Tendon of flexor digitorum superficialis **[T]**
d) Tendon of palmaris longus [F]
e) Ulnar artery [F]
Note: Palmaris longus passes superficial to the flexor retinaculum.

3. Anterior relations of right kidney
a) Fundus of the stomach [F]
b) 2nd part of duodenum **[T]**
c) Tail of pancreas [F]
d) Right suprarenal gland **[T]**
e) Right lobe of liver **[T]**

4. Muscles acting on both hip and knee
a) Gluteus maximus [F]
b) Sartorius **[T]**
c) Adductor longus [F]
d) Gracilis **[T]**
e) Popliteus [F]

5. Left coronary artery (left posterior aortic sinus)
a) Arises from ascending aorta **[T]**
b) Supplies the whole left atrium [F]
c) Runs in the anterior interventricular groove **[T]**
d) Gives off left circumflex artery **[T]**
e) Is a large artery [F]
Note: Orbit's key differs on c.

6. Muscles crossing shoulder joint
a) Brachialis [F]
b) Coracobrachialis **[T]**
c) Brachioradialis [F]
d) Biceps brachii **[T]**
e) Deltoid **[T]**

7. Pancreas
a) It is an endocrine gland **[T]**
b) It lies in the epigastrium **[T]**
c) It is an intra-peritoneal structure [F]
d) Alpha cells secrete insulin [F]
e) Portal vein is formed behind its neck **[T]**
Note: Alpha cells secrete glucagon, so d is F.

8. Parts of uterine tube
a) Isthmus **[T]**
b) Uterine **[T]**
c) Ovarian [F]
d) Ampulla **[T]**
e) Infundibulum **[T]**

9. Structures crossed by root of mesentery
a) Right ureter **[T]**
b) Inferior vena cava **[T]**
c) Left ureter [F]
d) Right psoas major muscle **[T]**
e) 2nd part of the duodenum [F]

10. Opening of right atrium
a) Superior vena cava **[T]**
b) Coronary sinus **[T]**
c) Anterior cardiac vein **[T]**
d) Pulmonary veins [F]
e) Aorta [F]

11. Structure passing through venacaval opening
a) Superior vena cava
>> b) Inferior vena cava
c) Thoracic duct
d) Gastric nerve
e) Azygos vein

12. NOT present in gastric gland
a) Chief cell
>> b) M cell
c) Parietal cell
d) Mucus neck cell
e) G cell

13. Retroperitoneal organ
a) Stomach
b) Spleen
>> c) Kidney
d) Transverse colon
e) Small intestine

14. Continuation of axillary artery
a) Subclavian artery
b) Lateral thoracic artery
>> c) Brachial artery
d) Subscapular artery
e) Circumflex humeral artery

15. Loss of knee extension, injury to
a) Sciatic nerve
b) Tibial nerve
>> c) Femoral nerve
d) Deep peroneal nerve
e) Superficial peroneal nerve

16. Deltoid muscle
>> a) Is a multipennate muscle
b) Is supplied by radial nerve
c) Forms the main bulk of back of arm
d) Has insertion on spinous process of scapula
e) Clavicular fibers are involved in abduction of shoulder joint

17. Duodenum developed from
a) Midgut
b) Foregut
>> c) Foregut and midgut
d) Hindgut
e) Pharyngeal gut

18. Commonest type of vermiform appendix
a) Paracolic
b) Subcaecal
>> c) Retrocaecal
d) Pelvic
e) Splenic

19. Ilio-tibial tract is condensation of
a) Femoral sheath
>> b) Deep fascia of thigh
c) Fascia iliaca
d) Quadriceps femoris muscle
e) Fascia cruris

20. SA node located at
a) Interatrial septum
b) Coronary sulcus
c) Fossa ovalis
>> d) Crista terminalis
e) Interventricular septum

November 2022 - Paper I (Orbit p.87)

1. Duodenum (stem misprinted 'DNA found in')
a) Lined by simple columnar epithelium **[T]**
b) Has numerous goblet cells **[T]**
c) Has Brunner's gland in submucosa **[T]**
d) Has leaf shaped villi **[T]**
e) Has Peyer's patches [F]
Note: The options describe the duodenum (compare Nov-2021 Paper II Q5). Orbit's key (only d T) does not match duodenum.

2. Gall bladder
a) Has capacity of 30-50 ml **[T]**
b) Stores bile **[T]**
c) Is supplied by hepatic artery **[T]**
d) Develops from midgut [F]
e) Drains by cystic duct **[T]**
Note: Cystic artery arises from right hepatic artery. Orbit's key differs between papers.

3. Structures passing through sacral canal
a) Sacral nerve **[T]**
b) Filum terminale **[T]**
c) Lower lumbar nerve [F]
d) Coccygeal nerve **[T]**
e) Obturator nerve [F]

4. Flexors of knee joint
a) Quadriceps femoris [F]
b) Semitendinosus **[T]**
c) Semimembranosus **[T]**
d) Biceps femoris **[T]**
e) Gracilis **[T]**
Note: Gracilis flexes and medially rotates the knee; Orbit marks it F.

5. Arch of aorta
a) Has three branches **[T]**
b) Is an elastic artery **[T]**
c) Lies in middle mediastinum [F]
d) Arises from left atrium [F]
e) Is a retroperitoneal structure [F]

6. Contents of spermatic cord
a) Vas deferens **[T]**
b) Testicular artery **[T]**
c) Artery to vas deferens **[T]**
d) Deep artery of penis [F]
e) Cremasteric fascia [F]
Note: Cremasteric fascia is a covering, not a content.

7. Porto-systemic anastomoses
a) Liver [F]
b) Oesophagus **[T]**
c) Anal canal **[T]**
d) Pancreas [F]
e) Caecum [F]

8. Parts of parietal pleura
a) Cervical **[T]**
b) Costal **[T]**
c) Mediastinal **[T]**
d) Diaphragmatic **[T]**
e) Peritoneal [F]
Note: Cervical pleura is a part (Orbit marks F). 'Peritoneal' is not pleura.

9. Structures crossed by root of mesentery
a) Abdominal aorta **[T]**
b) Left ureter [F]
c) Inferior vena cava **[T]**
d) Right psoas **[T]**
e) 2nd part of duodenum [F]

10. Branches of lumbar plexus
a) Genitofemoral nerve **[T]**
b) Femoral nerve **[T]**
c) Pudendal nerve [F]
d) Obturator nerve **[T]**
e) Nerve to pectineus [F]
Note: Nerve to pectineus comes from the femoral nerve (or accessory obturator); debatable. Pudendal is sacral plexus.

11. Nerves related to humerus
a) Axillary nerve **[T]**
b) Musculocutaneous nerve [F]
c) Ulnar nerve **[T]**
d) Radial nerve **[T]**
e) Long thoracic nerve [F]

12. Structures opening in right atrium
a) Superior vena cava **[T]**
b) Oblique vein of left ventricle [F]
c) Inferior vena cava **[T]**
d) Coronary sinus **[T]**
e) Great cardiac vein [F]

13. Branches of posterior cord of brachial plexus
a) Axillary nerve **[T]**
b) Thoracodorsal nerve **[T]**
c) Long thoracic nerve [F]
d) Radial nerve **[T]**
e) Upper and lower subscapular nerve **[T]**

14. Structures related to ala of sacrum
a) Obturator nerve **[T]**
b) Lumbosacral trunk **[T]**
c) Sympathetic trunk **[T]**
d) Iliolumbar artery **[T]**
e) Median sacral artery [F]
Note: Median sacral artery runs in the midline of the sacrum. Orbit's keys disagree between years.

15. Muscles supplied by musculocutaneous nerve
a) Biceps brachii **[T]**
b) Teres minor [F]
c) Coracobrachialis **[T]**
d) Subscapularis [F]
e) Brachialis **[T]**

16. Umbilical cord contains
a) Yolk sac with vitelline duct **[T]**
b) Allantois **[T]**
c) Two veins [F]
d) One artery [F]
e) Wharton's jelly **[T]**

17. Intracapsular structures of knee
a) Cruciate ligament **[T]**
b) Menisci **[T]**
c) Popliteal tendon **[T]**
d) Patellar ligament [F]
e) Synovial membrane [F]

18. Derivatives of paramesonephric duct
a) Uterus **[T]**
b) Lower part of vagina [F]
c) Fallopian tube **[T]**
d) Vas deferens [F]
e) Broad ligament of uterus [F]
Note: Uterus and tubes; broad ligament is a peritoneal fold; lower vagina is partly urogenital sinus.

19. Structures passing behind inguinal ligament
a) Femoral artery **[T]**
b) Femoral nerve **[T]**
c) Psoas major **[T]**
d) Femoral vein **[T]**
e) Spermatic cord [F]

20. Trachea
a) Lined by columnar epithelium **[T]**
b) Contains hyaline cartilage **[T]**
c) Divides at the level of T4 vertebra **[T]**
d) Contains goblet cells **[T]**
e) Contains C-shaped cartilage **[T]**
Note: Bifurcates at T4 (sternal angle level).

November 2022 - Paper II (Orbit p.88)

1. DNA found in
a) Centrosome [F]
b) Chromosome **[T]**
c) Lysosome [F]
d) Mitochondria **[T]**
e) Ribosome [F]

2. Simple squamous epithelium present in
a) Blood vessels **[T]**
b) Lung alveoli **[T]**
c) Pleura **[T]**
d) Tongue [F]
e) Urinary bladder [F]

3. Non-membrane-bound organelles
a) Centriole **[T]**
b) Lysosome [F]
c) Microtubules **[T]**
d) Mitochondria [F]
e) Ribosome **[T]**

4. Neural crest cell derivatives
a) Adrenal medulla **[T]**
b) Craniofacial skeleton **[T]**
c) Dorsal root ganglia of spinal nerve **[T]**
d) Epidermis [F]
e) Parotid gland [F]

5. Hyaline cartilage present in
a) Auditory tube [F]
b) Epiglottis [F]
c) Intervertebral disc [F]
d) Thyroid cartilage **[T]**
e) Trachea **[T]**

6. Fibrous joint
a) Coronal suture **[T]**
b) Inferior tibiofibular joint **[T]**
c) Intervertebral joint [F]
d) Manubriosternal joint [F]
e) Temporomandibular joint [F]

7. Central lymphoid organ
a) Bone marrow **[T]**
b) Lymph node [F]
c) Palatine tonsil [F]
d) Spleen [F]
e) Thymus **[T]**
Note: Bone marrow and thymus are central; spleen is peripheral. Orbit marks bone marrow F and spleen T - wrong.

8. Osteoblast
a) Is a bone forming cell **[T]**
b) Is a multinucleated cell [F]
c) Is a bone remodeling cell [F]
d) Is a mature cell [F]
e) Produces collagen fiber **[T]**

9. Pneumatic bones
a) Clavicle [F]
b) Frontal **[T]**
c) Maxillae **[T]**
d) Parietal [F]
e) Sphenoid **[T]**
Note: Frontal is pneumatic; Orbit marks it F.

10. Sperm
a) Always contains X chromosome [F]
b) Has got a tail **[T]**
c) Is a diploid cell [F]
d) Is a mature germ cell **[T]**
e) Is normally motile **[T]**

11. Carotid sheath contains
a) Common carotid artery **[T]**
b) Internal jugular vein **[T]**
c) Phrenic nerve [F]
d) Sympathetic trunk [F]
e) Vagus nerve **[T]**

12. Sensory cranial nerve
a) Facial [F]
b) Hypoglossal [F]
c) Olfactory **[T]**
d) Optic **[T]**
e) Trigeminal [F]

13. Branch of external carotid artery
a) Anterior cerebral [F]
b) Internal thoracic [F]
c) Lingual **[T]**
d) Superior thyroid **[T]**
e) Vertebral [F]

14. Derivatives of 1st pharyngeal arch
a) Medial pterygoid muscle **[T]**
b) Mylohyoid muscle **[T]**
c) Posterior belly of digastric muscle [F]
d) Stylopharyngeus muscle [F]
e) Temporalis muscle **[T]**

15. Muscles of soft palate
a) Genioglossus [F]
b) Levator palati **[T]**
c) Stylopharyngeus [F]
d) Tensor palati **[T]**
e) Tensor tympani [F]

16. Brainstem includes
a) Cerebellum [F]
b) Medulla oblongata **[T]**
c) Mid brain **[T]**
d) Pons **[T]**
e) Thalamus [F]

17. Parts of internal capsule
a) Anterior limb **[T]**
b) Fornix [F]
c) Genu **[T]**
d) Globus pallidus [F]
e) Splenium [F]

18. Deep cerebellar nuclei
a) Cuneatus [F]
b) Dentate **[T]**
c) Emboliform **[T]**
d) Fastigial **[T]**
e) Red nucleus [F]

19. Gray matter composed of
a) Blood vessels **[T]**
b) Cell body of neuron **[T]**
c) Fibroblast [F]
d) Lymphatics [F]
e) Myelinated nerve fiber [F]

20. Retina contains cells
a) Bipolar **[T]**
b) Ganglionic **[T]**
c) Purkinje [F]
d) Pyramidal [F]
e) Rod **[T]**

May 2022 - Paper I (Orbit p.89)

1. Mitotic cell division
a) Occurs in germ cells [F]
b) Occurs in somatic cells **[T]**
c) Is called reduction division [F]
d) Produce two daughter cells **[T]**
e) Causes genetic variation in daughter cells [F]

2. Oogenesis
a) Produce single haploid ovum **[T]**
b) Produce single polar body [F]
c) Requires estrogen hormone [F]
d) Starts before birth **[T]**
e) Starts from puberty [F]

3. Turner syndrome
a) Infertile **[T]**
b) Karyotype is 45X0 **[T]**
c) Karyotype is 47XXX [F]
d) Karyotype is 47XXY [F]
e) Tall stature [F]

4. Results of fertilization
a) Determination of chromosomal sex **[T]**
b) Initiation of cleavage **[T]**
c) Initiation of meiotic cell division [F]
d) Restoration of diploid chromosomes **[T]**
e) Restoration of haploid chromosomes [F]

5. Characteristics of epithelial tissue
a) Can regenerate **[T]**
b) Cellular component is more **[T]**
c) Covers only external surface of the body [F]
d) Has blood supply [F]
e) Has no nerve supply [F]
Note: Epithelium regenerates well, is cell-rich, covers external AND lines internal surfaces, is avascular, and has a nerve supply. Orbit's key (b to e T) is wrong on c, d, e.

6. Synovial joint
a) Ankle **[T]**
b) Costochondral [F]
c) Elbow **[T]**
d) Manubriosternal [F]
e) Shoulder **[T]**

7. Avascular structure
a) Bone [F]
b) Cornea **[T]**
c) Dermis [F]
d) Epidermis **[T]**
e) Hair **[T]**

8. Morphologically short bones
a) Calcaneum **[T]**
b) Capitate **[T]**
c) Cuboid **[T]**
d) Incus [F]
e) Patella [F]
Note: Incus (ear ossicle) is not clearly a short bone; Orbit marks it T. Patella is sesamoid.

9. Morula
a) Consists of eight blastomeres [F]
b) Consists of sixteen blastomeres **[T]**
c) Covered by zona pellucida **[T]**
d) Forms in uterine cavity [F]
e) Forms three days after fertilization **[T]**

10. Peripheral lymphoid organ
a) Bone marrow [F]
b) Lymph node **[T]**
c) Palatine tonsil **[T]**
d) Spleen **[T]**
e) Thymus [F]
Note: Orbit marks bone marrow T and spleen/thymus F - wrong.

11. Taste sensation from tongue carried by
a) Abducent nerve [F]
b) Accessory nerve [F]
c) Facial nerve **[T]**
d) Glossopharyngeal nerve **[T]**
e) Hypoglossal nerve [F]

12. Carotid triangle contains
a) Common carotid artery **[T]**
b) Hypoglossal nerve **[T]**
c) Internal jugular vein **[T]**
d) Submandibular gland [F]
e) Vagus nerve **[T]**

13. Mixed cranial nerve
a) Facial **[T]**
b) Glossopharyngeal **[T]**
c) Optic [F]
d) Oculomotor [F]
e) Trochlear [F]

14. Muscles of mastication
a) Lateral pterygoid **[T]**
b) Masseter **[T]**
c) Mylohyoid [F]
d) Styloglossus [F]
e) Temporalis **[T]**

15. Paired dural venous sinuses
a) Cavernous **[T]**
b) Occipital [F]
c) Straight [F]
d) Superior sagittal [F]
e) Transverse **[T]**

16. Hind brain includes
a) Cerebellum **[T]**
b) Cerebrum [F]
c) Medulla oblongata **[T]**
d) Midbrain [F]
e) Pons **[T]**

17. Cells in cerebral cortex
a) Ependymal cell [F]
b) Purkinje cell [F]
c) Pyramidal **[T]**
d) Schwann cell [F]
e) Stellate cell **[T]**

18. Sensory tracts of spinal cord
a) Corticospinal tract [F]
b) Rubrospinal tract [F]
c) Spinothalamic tract **[T]**
d) Tract of Burdach **[T]**
e) Tract of Goll **[T]**

19. Commissural fibers
a) Corpus callosum **[T]**
b) Habenular commissure **[T]**
c) Internal capsule [F]
d) Optic radiation [F]
e) Superior longitudinal fasciculus [F]

20. Muscle supplied by oculomotor nerve
a) Inferior oblique **[T]**
b) Inferior rectus **[T]**
c) Lateral rectus [F]
d) Medial rectus **[T]**
e) Superior oblique [F]

May 2022 - Paper II (Orbit p.90)

1. Muscles forming rotator cuff
a) Supraspinatus **[T]**
b) Subscapularis **[T]**
c) Infraspinatus **[T]**
d) Teres minor **[T]**
e) Teres major [F]

2. Structures related to neck of 1st rib
a) Sympathetic chain **[T]**
b) Subclavian vein [F]
c) 1st posterior intercostal vein **[T]**
d) Superior intercostal artery **[T]**
e) Phrenic nerve [F]

3. Fallopian tube
a) Lies in broad ligament **[T]**
b) Develops from mesonephric duct [F]
c) Has ampulla, the site of fertilization **[T]**
d) Is supplied by uterine and ovarian artery **[T]**
e) Is common site of ectopic pregnancy **[T]**

4. Muscles with double nerve supply
a) Deltoid [F]
b) Biceps brachii [F]
c) Brachialis **[T]**
d) Triceps [F]
e) Flexor digitorum profundus **[T]**

5. Structures related to transpyloric plane
a) Lower pole of kidney [F]
b) Fundus of gall bladder **[T]**
c) Bifurcation of abdominal aorta [F]
d) Pylorus of stomach **[T]**
e) Conus medullaris of spinal cord **[T]**

6. Gluteus maximus
a) Is attached to intertrochanteric line of femur [F]
b) Is extensor of hip joint **[T]**
c) Innervated by superior gluteal nerve [F]
d) Attached to iliotibial tract **[T]**
e) Attached to lesser trochanter of femur [F]

7. Fallot's tetralogy
a) Pulmonary stenosis **[T]**
b) Ventricular septal defect **[T]**
c) Atrial septal defect [F]
d) Overriding of aorta **[T]**
e) Left ventricular hypertrophy [F]

8. Structures crossing pelvic brim
a) Rectum [F]
b) Sigmoid colon **[T]**
c) Ureter **[T]**
d) Common iliac vessels **[T]**
e) Uterine tube [F]
Note: Ureter crosses at the bifurcation of the common iliac artery.

9. Portal triad contains
a) Branch of portal vein **[T]**
b) Branch of hepatic artery **[T]**
c) Bile canaliculi [F]
d) Interlobular hepatic ductule **[T]**
e) Central vein [F]
Note: Triad = portal venule, hepatic arteriole, bile ductule. Bile canaliculi and central vein are not part of it. Orbit's key (c T, d F) looks swapped.

10. Muscles causing dorsiflexion of foot
a) Tibialis anterior **[T]**
b) Extensor hallucis longus **[T]**
c) Extensor digitorum longus **[T]**
d) Tibialis posterior [F]
e) Peroneus longus [F]

11. The liver
a) Is a mixed gland **[T]**
b) Has bare area on superior surface [F]
c) Has central vein in hepatic lobule **[T]**
d) Has parenchyma develop from endoderm **[T]**
e) Has stroma develop from mesoderm **[T]**
Note: Bare area is on the posterior part of the diaphragmatic surface, not 'superior'.

12. The pancreas
a) Is mixed gland **[T]**
b) Is an intraperitoneal organ [F]
c) Has splenic vein on posterior groove **[T]**
d) Has parenchyma develop from endoderm **[T]**
e) Has duct opening in major duodenal papilla **[T]**

13. Structures related to base of heart
a) Descending aorta **[T]**
b) Vagus nerve [F]
c) Arch of aorta [F]
d) Phrenic nerve [F]
e) Thoracic duct **[T]**

14. The kidney
a) Is a retroperitoneal organ **[T]**
b) Has right kidney higher than that of left [F]
c) Develops from intermediate mesoderm **[T]**
d) Ascends from pelvic to lumbar region **[T]**
e) Has renal artery, a branch of iliac artery [F]

15. Ureter
a) Is 25 cm in length **[T]**
b) Crosses bifurcation of common iliac artery **[T]**
c) Has constriction opposite pelvi-ureteric junction **[T]**
d) Is lined by transitional epithelium **[T]**
e) Develops from endoderm [F]

16. Aortic opening of diaphragm transmits
a) Aorta **[T]**
b) Azygos vein **[T]**
c) Thoracic duct **[T]**
d) Sympathetic trunk [F]
e) Vagus nerve [F]

17. The spleen
a) Closely related to stomach **[T]**
b) Located deep to 9th-10th ribs [F]
c) Closely related to left kidney **[T]**
d) Site of production of erythrocyte after birth [F]
e) A less vascular organ [F]
Note: Spleen lies deep to ribs 9-11.

18. Contents of femoral sheath
a) Femoral nerve [F]
b) Femoral artery **[T]**
c) Genitofemoral nerve **[T]**
d) Femoral vein **[T]**
e) Lymph nodes **[T]**

19. Foot drop is due to injury to
a) Tibial [F]
b) Obturator [F]
c) Common peroneal **[T]**
d) Medial plantar [F]
e) Lateral plantar [F]
Note: Common peroneal nerve.

20. Conducting system of heart
a) SA node **[T]**
b) AV node **[T]**
c) Papillary muscle [F]
d) AV bundle **[T]**
e) Musculi pectinati [F]

November 2021 - Paper I (Orbit p.91)

1. Phagocytic cells
a) Macrophage **[T]**
b) Kupffer cell **[T]**
c) Fibroblast [F]
d) Osteocyte [F]
e) Microglia **[T]**
Note: Osteoclast (not osteocyte) is phagocytic; Orbit marks d T.

2. Meiotic type of cell division
a) Is called reduction division **[T]**
b) Occurs in germ cells **[T]**
c) Occurs in somatic cells [F]
d) Causes genetic variation in the daughter cells **[T]**
e) Produces two daughter cells [F]

3. Connective tissue cells
a) Fibroblast **[T]**
b) Schwann cell [F]
c) Plasma cell **[T]**
d) Astrocyte [F]
e) Goblet cell [F]

4. Neural crest derivatives
a) Hair [F]
b) Melanocyte **[T]**
c) Oligodendrocyte [F]
d) Dorsal root ganglia **[T]**
e) Schwann cell **[T]**

5. Ball and socket joint
a) Ankle joint [F]
b) Knee joint [F]
c) Hip joint **[T]**
d) Elbow joint [F]
e) Shoulder joint **[T]**

6. Pneumatic bones
a) Occipital [F]
b) Maxilla **[T]**
c) Frontal **[T]**
d) Zygomatic [F]
e) Temporal **[T]**
Note: Temporal bone (mastoid air cells) is pneumatic; Orbit marks it F.

7. Central lymphoid organs
a) Thymus **[T]**
b) Spleen [F]
c) Bone marrow **[T]**
d) Palatine tonsil [F]
e) Lymph node [F]

8. Fibrocartilage present in
a) Manubriosternal joint **[T]**
b) Elbow joint [F]
c) Intervertebral joint **[T]**
d) Symphysis pubis joint **[T]**
e) Inferior tibiofibular joint [F]
Note: IVD annulus fibrosus is fibrocartilage (Orbit marks c F). Inferior tibiofibular is a fibrous syndesmosis.

9. Collagen fibers abundant in
a) Bone **[T]**
b) Lymph node [F]
c) Lung [F]
d) Tendon **[T]**
e) Spleen [F]

10. Multipennate muscles
a) Soleus [F]
b) Deltoid **[T]**
c) Rectus femoris [F]
d) Tibialis anterior [F]
e) Subscapularis **[T]**
Note: Soleus and rectus femoris are bipennate. Orbit's key differs.

11. Motor cranial nerve
a) Hypoglossal **[T]**
b) Optic [F]
c) Trochlear **[T]**
d) Trigeminal [F]
e) Facial [F]

12. Derivatives of 1st pharyngeal arch
a) Mandible **[T]**
b) Stylopharyngeus muscle [F]
c) Palatine tonsil [F]
d) Temporalis muscle **[T]**
e) Tensor tympani muscle **[T]**

13. Branches of external carotid artery
a) Posterior auricular artery **[T]**
b) Anterior auricular artery [F]
c) Superior thyroid artery **[T]**
d) Inferior thyroid artery [F]
e) Lingual artery **[T]**
Note: Anterior auricular arteries arise from the superficial temporal (an ECA terminal branch) - indirect. Inferior thyroid is from thyrocervical trunk.

14. Parasympathetic fibers present in
a) Optic [F]
b) Oculomotor **[T]**
c) Facial **[T]**
d) Glossopharyngeal **[T]**
e) Accessory [F]
Note: Vagus also carries them.

15. Paired cartilage of larynx
a) Epiglottis [F]
b) Thyroid [F]
c) Corniculate **[T]**
d) Cricoid [F]
e) Arytenoid **[T]**

16. Gray matter composed of
a) Cell body of neuron **[T]**
b) Myelinated axons [F]
c) Capillaries **[T]**
d) Lymph vessels [F]
e) Neuroglial cells **[T]**

17. Brainstem includes
a) Mid brain **[T]**
b) Cerebellum [F]
c) Pons **[T]**
d) Thalamus [F]
e) Medulla oblongata **[T]**

18. Deep cerebellar nuclei
a) Dentate nucleus **[T]**
b) Red nucleus [F]
c) Fastigial nucleus **[T]**
d) Emboliform nucleus **[T]**
e) Caudate nucleus [F]

19. Parts of internal capsule
a) Rostrum [F]
b) Anterior limb **[T]**
c) Trunk [F]
d) Retrolentiform part **[T]**
e) Splenium [F]
Note: Rostrum, trunk and splenium are parts of the corpus callosum. Internal capsule: anterior limb, genu, posterior limb, retrolentiform, sublenticular.

20. Refractory media of eyeball
a) Iris [F]
b) Cornea **[T]**
c) Retina [F]
d) Lens **[T]**
e) Vitreous humor **[T]**

November 2021 - Paper II (Orbit p.92)

1. Gall bladder
a) Has capacity of 30-50 ml **[T]**
b) Stores bile **[T]**
c) Is supplied by hepatic artery **[T]**
d) Develops from midgut [F]
e) Drains by cystic duct **[T]**

2. Intercostal nerves
a) Are mixed nerves **[T]**
b) Are ventral rami of thoracic spinal nerves **[T]**
c) Supplies the diaphragm **[T]**
d) Supplies parietal pleura **[T]**
e) Supplies fibrous pericardium [F]
Note: Lower intercostals supply the peripheral diaphragm; fibrous pericardium is supplied by the phrenic nerve.

3. Lesser omentum contains
a) Hepatic artery **[T]**
b) Common bile duct **[T]**
c) Portal vein **[T]**
d) Inferior vena cava [F]
e) Coeliac trunk [F]
Note: Free border: hepatic artery proper, bile duct, portal vein. IVC and coeliac trunk are not contents.

4. Structures opening in right atrium
a) Superior vena cava **[T]**
b) Oblique vein of left ventricle [F]
c) Inferior vena cava **[T]**
d) Coronary sinus **[T]**
e) Great cardiac vein [F]

5. Histological structures of duodenum
a) Lined by simple columnar epithelium **[T]**
b) Has numerous goblet cells **[T]**
c) Has Brunner's gland in submucosa **[T]**
d) Has leaf shaped villi **[T]**
e) Has Peyer's patches [F]
Note: Peyer's patches are ileal.

6. Structure passing through sacral canal
a) Sacral nerve **[T]**
b) Filum terminale **[T]**
c) Lower lumbar nerve [F]
d) Coccygeal nerve **[T]**
e) Obturator nerve [F]

7. Structures related to ala of sacrum
a) Obturator nerve **[T]**
b) Lumbosacral trunk **[T]**
c) Sympathetic trunk **[T]**
d) Iliolumbar artery **[T]**
e) Median sacral artery [F]
Note: See note in Nov-2022 Paper I.

8. Branches of lumbar plexus
a) Genitofemoral nerve **[T]**
b) Femoral nerve **[T]**
c) Pudendal nerve [F]
d) Obturator nerve **[T]**
e) Nerve to pectineus [F]

9. Contents of spermatic cord
a) Vas deferens **[T]**
b) Testicular artery **[T]**
c) Artery to vas deferens **[T]**
d) Deep artery of penis [F]
e) Cremasteric fascia [F]

10. Derivatives of paramesonephric duct
a) Uterus **[T]**
b) Lower part of vagina [F]
c) Fallopian tube **[T]**
d) Vas deferens [F]
e) Round ligament of uterus [F]
Note: Round ligament of uterus comes from the gubernaculum; Orbit marks e T.

11. Nerves related to humerus
a) Axillary nerve **[T]**
b) Musculocutaneous nerve [F]
c) Ulnar nerve **[T]**
d) Radial nerve **[T]**
e) Long thoracic nerve [F]

12. Common congenital anomalies of foot
a) Club foot **[T]**
b) Flat foot [F]
c) Talipes equinovarus **[T]**
d) Syndactyly **[T]**
e) Talipes equinovalgus [F]

13. Intracapsular structures of knee joint
a) Cruciate ligament **[T]**
b) Menisci **[T]**
c) Popliteal tendon **[T]**
d) Patellar ligament [F]
e) Synovial membrane [F]

14. Structures passing behind inguinal ligament
a) Femoral artery **[T]**
b) Femoral nerve **[T]**
c) Psoas major **[T]**
d) Femoral vein **[T]**
e) Spermatic cord [F]

15. Radial nerve supplies
a) Biceps brachii [F]
b) Anconeus **[T]**
c) Extensor carpi radialis longus **[T]**
d) Extensor carpi ulnaris **[T]**
e) Brachioradialis **[T]**
Note: Anconeus is supplied by the radial nerve (Orbit marks b F).

16. The clavicle
a) Is a modified long bone **[T]**
b) Lateral end articulates with manubrium [F]
c) It ossifies in membrane **[T]**
d) Deltoid attached to medial two-thirds of it [F]
e) It has no medullary cavity **[T]**

17. Muscles innervated by femoral nerve
a) Rectus femoris **[T]**
b) Biceps femoris [F]
c) Sartorius **[T]**
d) Adductor longus [F]
e) Vastus lateralis **[T]**

18. Venacaval opening of the diaphragm
a) Transmits inferior vena cava **[T]**
b) Is a tendinous opening **[T]**
c) Transmits left phrenic nerve [F]
d) Transmits vagus nerve [F]
e) Lies at the level of T10 vertebra [F]
Note: At T8 (esophageal opening T10, aortic T12). Left phrenic pierces the muscle.

19. The female pelvis
a) Inlet is heart shaped [F]
b) Ischial spine is less inverted **[T]**
c) Cavity wider and shallower **[T]**
d) Cavity is narrower [F]
e) Subpubic angle wider **[T]**

20. Invertors of foot
a) Extensor digitorum [F]
b) Tibialis anterior **[T]**
c) Peroneus brevis [F]
d) Extensor hallucis longus [F]
e) Tibialis posterior **[T]**
Note: EHL is mainly a dorsiflexor.

May 2021 - Paper I (Orbit p.93)

1. Membranous organelles
a) Lysosomes **[T]**
b) Endoplasmic reticulum **[T]**
c) Mitochondria **[T]**
d) Microfilaments [F]
e) Ribosomes [F]

2. Mitotic cell division occurs in
a) Primary oocyte [F]
b) Zygote **[T]**
c) Somatic cells **[T]**
d) Spermatozoa [F]
e) Oogonia **[T]**

3. Surface ectodermal derivatives
a) Hair **[T]**
b) Dermis [F]
c) Thyroid gland [F]
d) Sweat gland **[T]**
e) Bone [F]

4. Primary cartilaginous joints
a) Intervertebral joint [F]
b) 1st sternocostal joint **[T]**
c) Manubriosternal joint [F]
d) Temporomandibular joint [F]
e) Spheno-occipital joint **[T]**

5. Paranasal air sinuses
a) Frontal **[T]**
b) Maxillary **[T]**
c) Cavernous [F]
d) Sphenoidal **[T]**
e) Occipital [F]

6. Surface modifications of cell
a) Microvilli **[T]**
b) Cilia **[T]**
c) Villi [F]
d) Papillae [F]
e) Flagella **[T]**
Note: Flagella are counted as a surface modification (as in May-2023); Orbit marks it F here.

7. Elastic cartilage present in
a) Epiglottis **[T]**
b) Thyroid cartilage [F]
c) Intervertebral disc [F]
d) External ear **[T]**
e) Articular cartilage [F]

8. Peripheral lymphoid organs
a) Palatine tonsil **[T]**
b) Lymph node **[T]**
c) Bone marrow [F]
d) Thymus [F]
e) Spleen **[T]**

9. Functional end artery present in
a) Liver [F]
b) Heart **[T]**
c) Lung [F]
d) Brain **[T]**
e) Retina **[T]**

10. Osteoblast
a) Multinucleated cell [F]
b) Lies in lacunae of Haversian system [F]
c) Bone forming cell **[T]**
d) Converts into osteocyte **[T]**
e) Immature cell **[T]**

11. Sensory cranial nerve
a) Optic **[T]**
b) Facial [F]
c) Olfactory **[T]**
d) Trigeminal [F]
e) Glossopharyngeal [F]

12. Contents of carotid sheath
a) Phrenic nerve [F]
b) Common carotid artery **[T]**
c) Internal jugular vein **[T]**
d) Sympathetic trunk [F]
e) Vagus nerve **[T]**

13. Muscles of mastication
a) Temporalis **[T]**
b) Mentalis [F]
c) Genioglossus [F]
d) Stylohyoid [F]
e) Masseter **[T]**

14. Oropharynx
a) Contains palatine tonsil **[T]**
b) Contains tubal tonsil [F]
c) Is the middle part of the pharynx **[T]**
d) Is collapsible [F]
e) Communicates anteriorly with nasal cavity [F]

15. Muscles of soft palate
a) Tensor tympani [F]
b) Genioglossus [F]
c) Tensor veli palatini **[T]**
d) Stylohyoid [F]
e) Palatopharyngeus **[T]**

16. Frontal lobe of cerebrum contains
a) Primary sensory area [F]
b) Primary motor area **[T]**
c) Broca's area **[T]**
d) Frontal eye field **[T]**
e) Visual area [F]

17. Hind brain formed by
a) Medulla oblongata **[T]**
b) Thalamus [F]
c) Cerebellum **[T]**
d) Midbrain [F]
e) Hypothalamus [F]

18. Basal ganglia includes
a) Caudate nucleus **[T]**
b) Corpus callosum [F]
c) Globus pallidus **[T]**
d) Red nucleus [F]
e) Putamen **[T]**

19. Effects of cerebellar lesion
a) Chorea [F]
b) Dysdiadochokinesia **[T]**
c) Spastic paralysis [F]
d) Ataxia **[T]**
e) Nystagmus **[T]**

20. Oculomotor nerve innervates
a) Superior rectus **[T]**
b) Superior oblique [F]
c) Lateral rectus [F]
d) Medial rectus **[T]**
e) Inferior oblique **[T]**

May 2021 - Paper II (Orbit p.94)

1. The liver
a) Is a mixed gland **[T]**
b) Has bare area on superior surface [F]
c) Has central vein in hepatic lobule **[T]**
d) Has parenchyma develop from endoderm **[T]**
e) Has stroma develop from mesoderm **[T]**
Note: Bare area is on the posterior part of the diaphragmatic surface.

2. Structures crossed by root of mesentery
a) Abdominal aorta **[T]**
b) Left ureter [F]
c) Inferior vena cava **[T]**
d) Left psoas [F]
e) 2nd part of duodenum [F]

3. Hindgut derivatives
a) Rectum **[T]**
b) Anal canal below pectinate line [F]
c) Right 2/3 of transverse colon [F]
d) Ascending colon [F]
e) Vermiform appendix [F]

4. Structures opening in right atrium
a) Superior vena cava **[T]**
b) Oblique vein of left ventricle [F]
c) Inferior vena cava **[T]**
d) Coronary sinus **[T]**
e) Great cardiac vein [F]

5. Flexors of knee joint
a) Quadriceps femoris [F]
b) Semitendinosus **[T]**
c) Semimembranosus **[T]**
d) Biceps femoris **[T]**
e) Gracilis **[T]**

6. Structures passing through sacral canal
a) Sacral nerve **[T]**
b) Filum terminale **[T]**
c) Lower lumbar nerve [F]
d) Coccygeal nerve **[T]**
e) Obturator nerve [F]
Note: Orbit marks obturator nerve T - wrong.

7. The kidney
a) Is a retroperitoneal organ **[T]**
b) Has right kidney higher than left [F]
c) Develops from intermediate mesoderm **[T]**
d) Ascends from pelvic to lumbar region **[T]**
e) Has renal artery, a branch of iliac artery [F]

8. Claw hand injury involves
a) Median nerve [F]
b) Ulnar nerve **[T]**
c) Thoracic duct (sic) [F]
d) Axillary nerve [F]
e) Long thoracic nerve [F]
Note: Ulnar nerve. Printed 'thoracic duct' is unrelated (maybe 'thoracic outlet' misprint); Orbit marks c T.

9. The fallopian tube
a) Lies in broad ligament **[T]**
b) Develops from mesonephric duct [F]
c) Has ampulla, the site of fertilization **[T]**
d) Is supplied by uterine and ovarian artery **[T]**
e) Is common site of ectopic pregnancy **[T]**

10. Umbilical cord contains
a) Yolk sac with vitelline duct **[T]**
b) Allantois **[T]**
c) Two veins [F]
d) One artery [F]
e) Wharton's jelly **[T]**

11. Histological structures of duodenum
a) Lined by simple columnar epithelium **[T]**
b) Has numerous goblet cells **[T]**
c) Has Brunner's gland in submucosa **[T]**
d) Has leaf shaped villi **[T]**
e) Has Peyer's patches [F]

12. Arch of the aorta
a) Has three branches **[T]**
b) Is an elastic artery **[T]**
c) Lies in middle mediastinum [F]
d) Arises from left atrium [F]
e) Is a retroperitoneal structure [F]

13. Intracapsular structures of knee joint
a) Cruciate ligament **[T]**
b) Menisci **[T]**
c) Popliteal tendon **[T]**
d) Patellar ligament [F]
e) Synovial membrane [F]
Note: Synovial membrane lines the capsule; Orbit's key varies between years.

14. Structures passing behind inguinal ligament
a) Femoral artery **[T]**
b) Femoral nerve **[T]**
c) Psoas major **[T]**
d) Femoral vein **[T]**
e) Spermatic cord [F]

15. Tributaries of superior vena cava

a) Azygos vein **[T]**
b) Bronchial vein [F]
c) Hemiazygos vein [F]
d) Pericardial vein **[T]**
e) Brachiocephalic vein **[T]**

Note: Hemiazygos drains into the azygos vein, not directly into SVC. Small pericardial/mediastinal veins do drain directly.

16. Structures related to transpyloric plane

a) Lower pole of kidney [F]
b) Fundus of gall bladder **[T]**
c) Abdominal aorta bifurcation [F]
d) Pylorus of stomach **[T]**
e) Conus medullaris of spinal cord **[T]**

17. Muscles supplied by obturator nerve

a) Adductor magnus **[T]**
b) Biceps femoris [F]
c) Rectus femoris [F]
d) Adductor longus **[T]**
e) Gracilis **[T]**

18. Aortic opening of diaphragm transmits

a) Aorta **[T]**
b) Azygos vein **[T]**
c) Radial nerve [F]
d) Vagus nerve [F]
e) Sympathetic trunk [F]

Note: Thoracic duct also passes through it.

19. The female pelvis

a) Inlet is heart shaped [F]
b) Ischial spine is less inverted **[T]**
c) Cavity wider and shallower **[T]**
d) Cavity is narrower [F]
e) Subpubic angle wider **[T]**

20. Contents of spermatic cord

a) Vas deferens **[T]**
b) Testicular artery **[T]**
c) Artery to vas deferens **[T]**
d) Deep artery of penis [F]
e) Cremasteric fascia [F]